

Tixora Fraud Detection — Training Report

Best model: **xgboost** | Test PR-AUC: 0.825 | Test ROC-AUC: 0.963

Core-features-only input retains 99.1% of full-feature PR-AUC — deployable today without any application changes.

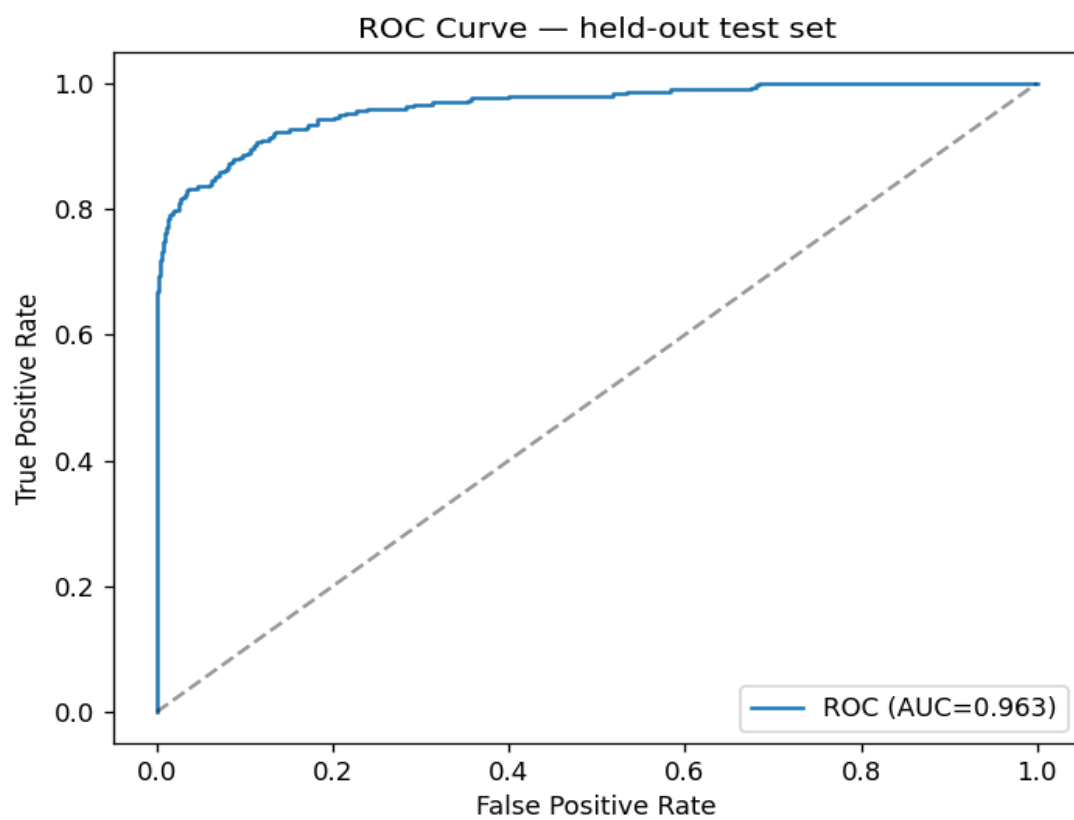
Model Comparison (validation PR-AUC)

model	val_pr_auc	train_seconds	tuned
xgboost	0.7926	99.9546	True
hist_gradient_boosting	0.7837	99.569	True
lightgbm	0.7697	2.1885	False
catboost	0.7693	5.0213	False
random_forest	0.7635	19.6804	False
extra_trees	0.6719	17.106	False
logistic_regression	0.5762	0.6929	False

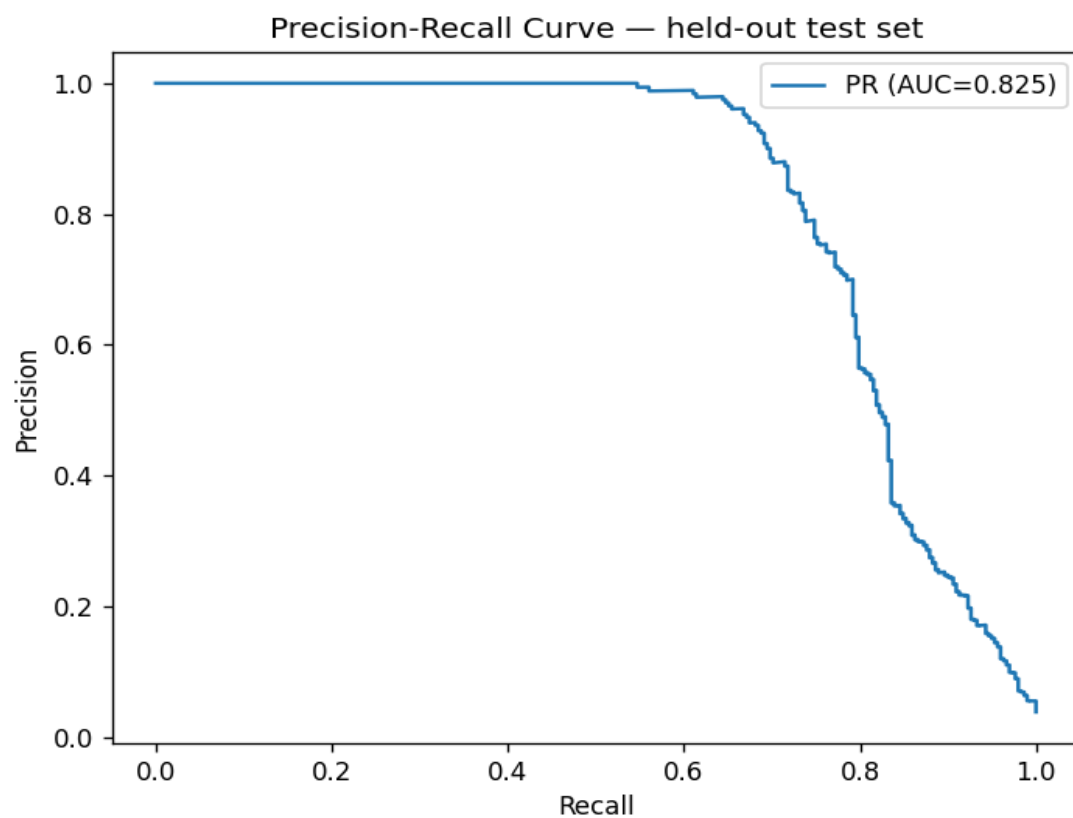
Held-out Test Set Metrics

Metric	Value
accuracy	0.9468
precision	0.4069
recall	0.8356
f1	0.5473
roc_auc	0.9635
pr_auc	0.8254

ROC Curve



Precision-Recall Curve



Top Feature Importances

